



Stabilization of formamidinium lead triiodide α -phase with isopropylammonium chloride for perovskite solar cells

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Formamidinium lead triiodide (FAPbI₃) perovskite solar cells (PSCs) are mainly fabricated by sequentially coating lead iodide and formamidinium iodide, or by coating a solution in which all components are dissolved in one solvent (one-pot process). The PSCs produced by both processes exhibited similar efficiencies; however, their long-term stabilities were notably different. We concluded that the major reason for this behaviour is the stabilization of the α -FAPbI₃ phase by isopropylammonium cations produced by the chemical reaction between isopropyl alcohol, used as solvent, and methylammonium chloride, added during the process. On this basis, we fabricated PSCs by adding isopropylammonium chloride to the perovskite precursor solution for the one-pot process and achieved a certified power conversion efficiency of 23.9%. Long-term operational current density–voltage measurements (one sweep every 84 min under 1-Sun irradiation in nitrogen atmosphere) showed that the as-fabricated device with an initial efficiency of approximately 20% recorded an efficiency of about 23% after 1,000 h that gradually degraded to about 22% after an additional 1,000 h.

Halide perovskite light absorbers of chemical formula APbI₃ (A represents a cation) exhibit typical three-dimensional (3D) structures. Formamidinium (FA, NH₂CH=NH₂⁺), methylammonium (MA, CH₃NH₃⁺) and caesium (Cs⁺) are mainly used as A-site cations, and they strongly affect the performance and stability of a solar device¹. Of these cations, FA shows the most promising performance because 3D formamidinium lead triiodide (FAPbI₃) perovskite has a lower bandgap (E_g) and is thermally more stable than its MA counterpart^{2,3}. However, 3D FAPbI₃ perovskite (α -phase) is metastable and undergoes phase conversion into a one-dimensional non-perovskite (δ -phase) at room temperature⁴. Several methods for stabilizing the α -phase of FAPbI₃ have been reported⁵. Generally, FAPbI₃ perovskite films obtained by solution processes are prepared by one-pot^{1–4,6} or sequential^{7,8} spin-coating methods followed by an annealing process to improve crystallization. The one-pot method usually entails anti-solvent treatment to form a dense FAPbI₃ film. Amongst the solvents used in the dropping process, isopropyl alcohol (IPA) enhances the crystal quality of the FAPbI₃ layer in the one-pot process⁹. The sequential process exploits the reaction between organic cation solutions dissolved in IPA and those predeposited in the PbI₂ layer.

In both the one-pot and sequential processes, additive engineering is an effective strategy for mediating perovskite crystal growth and passivating defect states. In particular, methylammonium chloride (MACl) additives have been used to improve the crystallinity and surface morphology of FAPbI₃ (ref. ⁵). However, the impact of the differences in FAPbI₃ perovskite crystallization on the efficiency and stability of the final perovskite solar cells (PSCs) has only been examined in a few cases¹⁰. Moreover, forming a high-quality stabilized α -phase FAPbI₃ perovskite film with large grain size, pre-

ferred crystal orientation and low trap density, which facilitates low non-radiative energy loss and long-term stability, is challenging.

In this study, we explored the differences in the FAPbI₃ perovskite crystals prepared by the one-pot and sequential spin-coating methods by in situ synchrotron-based grazing incidence wide-angle X-ray diffraction (GI-WAXD), high-resolution transmittance electron microscopy (HR-TEM), time-of-flight secondary ion mass spectroscopy (ToF-SIMS) and high-resolution X-ray photoelectron spectroscopy (HR-XPS). Based on the analytical results, we propose the inclusion of isopropylammonium cations (iPAmH⁺), mainly located at the grain boundaries of perovskite layers, formed by a secondary chemical side reaction, such as between MACl and IPA, in the sequential spin-coating process. In model experiments we fabricated PSC devices using a one-pot process in which 15 mol% isopropylammonium chloride (iPAmHCl) was added to the FAPbI₃ precursor solution. A substantial amount of the introduced iPAmHCl volatilizes during the annealing of the FAPbI₃ thin film at 150°C, with iPAmHCl remaining at the FAPbI₃ grain boundaries contributing to the stabilization of the α -phase, enabling PSCs with high efficiency and stability to be fabricated.

Comparison between the two processes using MACl and IPA

The PSCs were prepared by two representative methods: the one-pot and sequential processes. As shown in Supplementary Fig. 1a, no important difference was noted between the two processes in terms of efficiency. Because other parameters such as the device structure and charge carriers were the same, the similar efficiency implies that the properties of the perovskite films are similar. However, the sequentially processed PSC device showed better stability

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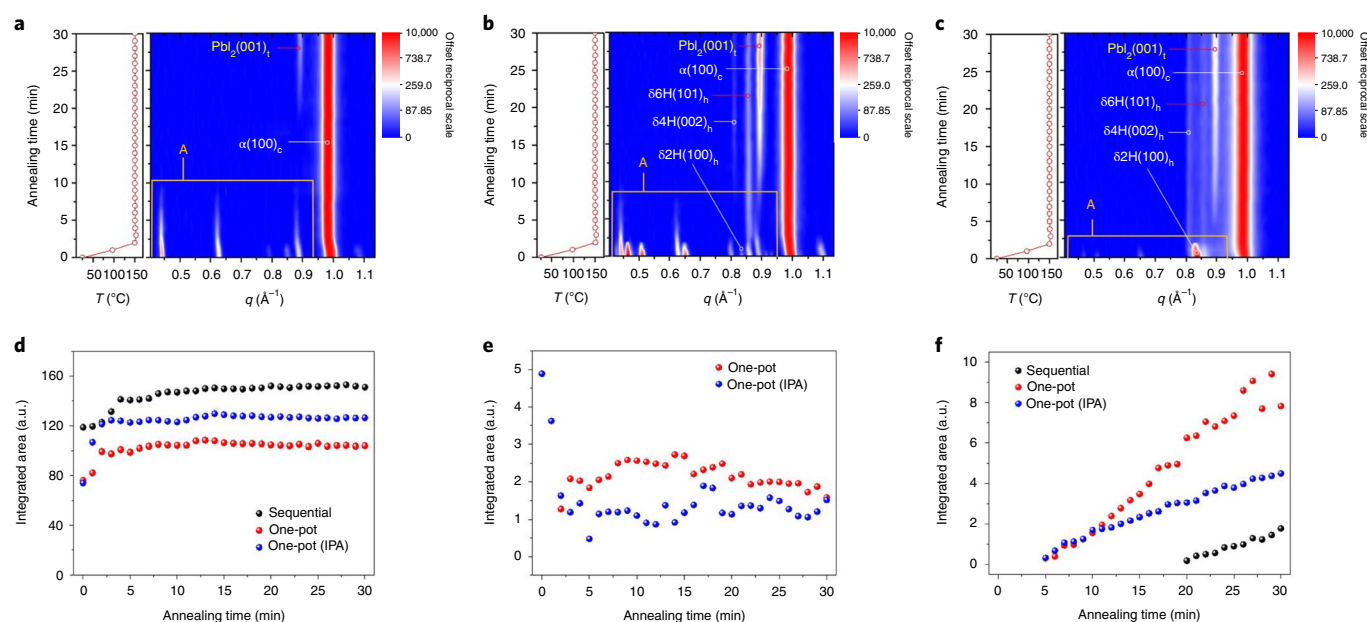


Fig. 1 | Monitoring the formation of FAPbI₃ films. a–c, Azimuthally integrated GI-WAXD patterns in the region between 0.30 and 1.15 Å⁻¹ of the as-deposited layers formed by the sequential (**a**), one-pot (**b**) and one-pot (IPA) (**c**) processes. The temperature–time profiles for the annealing process for each in situ experiment are shown on the left side of **a–c**. All the pristine samples showed many diffraction peaks in the A region of **a–c**, which are due to the formation of intermediate phases in the form of (solvent or additive)_a(FA or MA)_bPbI_d. Colour scale bars showing the offset reciprocal scale are shown on the right side of **a–c**. **d–f**, Integrated intensities of α -FAPbI₃ (100)_{cubic} (**d**), δ 4H-FAPbI₃ (002)_{hexagonal} + δ 6H-FAPbI₃ (101)_{hexagonal} (**e**) and PbI₂ (001)_{trigonal} (**f**). The integrated areas of **d**, **e** and **f** were extracted from **a**, **b** and **c**, respectively, by numerical integration using the equation $\int_{x_1}^{x_n} f(x)dx \approx \sum_{i=1}^{n-1} (x_{i+1} - x_i) \frac{1}{2} [f(x_{i+1}) + f(x_i)]$.

than that fabricated by the one-pot process (Supplementary Fig. 1b). As the perovskite thin films produced by both processes have almost the same thickness (Supplementary Fig. 2a), surface morphology (Supplementary Fig. 2b), light absorbance and bandgap (Supplementary Fig. 3a), and Cl⁻ concentration (Supplementary Fig. 3b), we can surmise that other factors stabilize the sequentially produced FAPbI₃ thin film. Regarding the chemical components, the only difference in the as-deposited layers formed by the two different processes is that IPA was used to dissolve the organic ammonium cations in the sequential process. To determine whether the inclusion of IPA affects the stability of the PSCs fabricated by the two processes, a comparison was made by adding IPA to a one-pot solution. We observed that the stability of the device fabricated by adding 20 mol% IPA to the one-pot solution notably increased compared with that fabricated without IPA, as shown in Supplementary Fig. 1c.

To understand the two different processes and the effect of the addition of IPA to the one-pot solution, the formation of as-deposited layers coated by the one-pot and sequential processes was monitored by in situ synchrotron GI-WAXD during thermal annealing. The one-pot solution with the addition of IPA is denoted as ‘one-pot (IPA)’. Figures 1a–c present the azimuthally integrated GI-WAXD patterns converted from the temperature- and time-dependent in situ GI-WAXD analyses. Representative two-dimensional (2D) GI-WAXD patterns are shown in Supplementary Fig. 4. All three processes clearly showed a strong α -FAPbI₃ (100)_{cubic} diffraction peak, denoted as α (100)_c, at $q = 0.9893$ Å⁻¹ (sequential), 0.9959 Å⁻¹ (one-pot) and 0.9888 Å⁻¹ (one-pot (IPA)) in the initial pristine thin films. Note that MACl acts as an important additive for the formation of α (100)_c in the as-deposited layers^{2,3}, which is evident from a comparison with a film prepared by the one-pot process without the MACl additive (Supplementary Fig. 5a). All the pristine samples also showed many diffraction peaks in the A region, attributed

to intermediate phases in the form of (solvent or additive)_a(FA or MA)_bPbI_d (refs. ^{11,12}), with the one-pot (IPA) and one-pot processes showing the smallest and largest number of diffraction peaks, respectively. This suggests that the added IPA inhibits the formation of complex intermediate phases and contributes to the conversion to the crystalline FAPbI₃ phase. The one-pot and one-pot (IPA) processes showed extremely weak or clear δ -FAPbI₃ diffractions of δ 4H(002)_h, δ 2H(100)_h and δ 6H(101)_h (where the subscript h stands for hexagonal) at $q = 0.811$, 0.839 and 0.866 Å⁻¹, respectively (Fig. 1b,c and Supplementary Fig. 5a,b)¹³, whereas no δ 2H(100)_h diffraction was observed in the sequential process (Fig. 1a). To examine the differences in α (100)_c crystal formation after annealing at 150 °C, the mutually normalized integrated areas of α (100)_c, δ 4H(002)_h, δ 6H(101)_h and PbI₂ (001)_{trigonal}, denoted as PbI₂(001)_i, for different annealing times were calculated (Fig. 1d–f). The integrated areas of α (100)_c reveal that more than 60% of α (100)_c was obtained in the beginning for all three samples and reached nearly 100% within 15 min of annealing at 150 °C (Fig. 1d). The ratio of the relative integrated areas of α (100)_c for one-pot/one-pot (IPA)/sequential is 1:1.2:1.5, which implies that the sequential process provides the most crystalline film. The integrated area of non-perovskite δ (100)_h shows the opposite tendency to that of α (100)_c (Fig. 1e), with δ (100)_h of the sequential process disappearing after 3 min. The FAPbI₃ film prepared by the one-pot process did not fully convert from δ (100)_h to α (100)_c crystals, whereas the one-pot (IPA) achieved an almost complete transformation. The thermal stability of the FAPbI₃ films can be evaluated by observing the changes in the integrated area of PbI₂(001)_i (Fig. 1f). The integrated area of PbI₂(100)_i of the one-pot (IPA) process showed a slower increase with annealing time at 150 °C than that of the one-pot process, indicating that the one-pot (IPA) process shows better thermal stability than the one-pot process. Moreover, the film prepared by the one-pot (IPA) showed a similar increase to that of the sequential

process after annealing for 20 min, suggesting a similar thermal stability after a certain annealing time. The thermal stability can also be predicted by monitoring the variation in the relative amount of $\alpha(100)_c$ with annealing time with respect to the total amount of $\alpha(100)_c$, $\delta(100)_h$ and $\text{PbI}_2(001)_i$ (Supplementary Fig. 5c). Initially, the sequentially processed film was highly stable, regardless of the annealing time; however, after annealing for approximately 25 min, the decrease in slope in the sequential and one-pot (IPA) samples was almost the same. In contrast, the film from the one-pot process tended to decrease sharply after annealing for 5 min. This is consistent with the higher stability of the PSCs fabricated by the sequential and one-pot (IPA) processes (Supplementary Fig. 1). Moreover, the development of crystal orientation over time was also observed to differ for each process (Supplementary Fig. 6).

Although the presence of IPA influences the crystallization of FAPbI_3 , the volatile solvent IPA does not have a important effect on the α -phase stabilization of the FAPbI_3 perovskite film. Thus, the structures of the FAPbI_3 thin films prepared by sequential, one-pot and one-pot (IPA) processes at 150 °C were investigated using HR-TEM. Representative cross-sectional HR-TEM images are shown in Fig. 2a–c, and the magnifications of selected areas, shown by the yellow solid-line squares, are displayed in Fig. 2d–f. Two distinct crystal structures can be observed inside and outside the yellow dashed ellipses. The interplanar spacings (ISs) of selected crystals in Fig. 2d–f were analysed and their d spacings and Miller indices (Supplementary Table 1 and Supplementary Fig. 7), together with their magnified HR-TEM images, are compared in Fig. 2g–i and Supplementary Fig. 8. For the FAPbI_3 film obtained by the sequential process, the crystals in the inner regions of the ellipses (IREs) in Fig. 2d, with an IS of 0.66 nm and a defined fast Fourier transform (FFT) pattern, were identified as the typical hexagonal PbI_2 domain. The FAPbI_3 crystals in the outer regions of the ellipses (OREs) appear as a matrix structure in Fig. 2d,g and Supplementary Fig. 8a; the IS values of 0.36 and 0.31 nm are in good agreement with $\alpha(111)$ and $\alpha(200)$ domains, respectively. For the FAPbI_3 layers formed by the one-pot process, the IS values of 0.33–0.35 and 0.24 nm in the IREs shown in Fig. 2e,h and Supplementary Fig. 8b exhibit good agreement with $\alpha(111)$ and $\text{PbI}_2(003)$ domains, respectively. The OREs of the FAPbI_3 layers formed by the one-pot (IPA) process (Fig. 2f) exhibit α - FAPbI_3 domains and amorphous phases. In the IREs in Fig. 2f,i and Supplementary Fig. 8c, the IS of 0.69 nm and the FFT pattern are almost identical to those of the PbI_2 domain in the IREs of the sequentially processed FAPbI_3 layer. In the OREs of Fig. 2f,i and Supplementary Fig. 8c, the crystal structure and FFT patterns were identified as the $\alpha(111)$ domain, and are highly similar to those of the sequentially processed FAPbI_3 crystal, as shown in Fig. 2g,i and Supplementary Fig. 8c. Here, we recognize that introducing IPA substantially contributes towards the growth of 5–15-nm-sized cubic α - FAPbI_3 domains and the absence of amorphous domains, as compared with those in the one-pot FAPbI_3 layer. This implies that the final FAPbI_3 crystal properties are affected by the addition of IPA, similarly to the different crystallization behaviour observed during the annealing of the as-deposited layers. IPA was not completely removed by volatilization during the heat treatment step, suggesting that some is converted into other materials, which might affect the stabilization of the perovskite crystals.

Transamination of IPA with MACl for iPA_mHCl in perovskites

The chemical species inside or outside the FAPbI_3 crystals deposited by the sequential, one-pot and one-pot (IPA) processes were investigated more closely using ToF-SIMS. This technology is one of the few that reveal chemical information on all components of halide perovskites^{5,14,15}. Three representative elements, namely cationic lead (Pb^+) with positive polarity and anionic iodide (I^-) and carbon (C_n^- , $n=1$ and 2) with negative polarity, were selected

for analysis because they are the basic components of the FAPbI_3 film. Supplementary Fig. 9a–d shows the calculated integrated populations of each element obtained from scanning ToF-SIMS analysis (Supplementary Table 2). The calculated concentrations of Pb and I were similar for all three FAPbI_3 films. However, the total carbon (C^- and C_2^-) concentrations in the sequential and one-pot (IPA) perovskite layers were higher (109.4 and 104.6%, respectively) than that observed in the one-pot film. This means that more organic materials were present in the FAPbI_3 thin films prepared by the processes including IPA.

Nevertheless, it was difficult to specify the organic compounds based on this analysis. If any inclusions exist inside or outside the FAPbI_3 crystal, the chemical binding energy or the position of the valence band of FAPbI_3 can vary slightly. Figure 3 shows the synchrotron-assisted HR-XPS spectra for C 1s and N 1s for the three perovskite films prepared by the sequential, one-pot and one-pot (IPA) processes. The spectra obtained were deconvoluted using the Voigt function to identify the precise components¹⁶. First, we analysed the C 1s spectra in Fig. 3a–c, which indicate five components. Four of the components were well assigned to C–C(or CH)¹⁷, C–N¹⁷, FA¹⁸ and C–I, in which I is bonded with C in MA or methane (marked as MA(complexes))¹⁷. A mismatching chemical component is present in the region 286.5–287.0 eV in the C 1s spectra of each of the three perovskite films. Of the three unknown peaks, the two in the sequential and one-pot (IPA) spectra exhibit a highly similar energy at ~286.4 eV with a full-width at half-maximum (FWHM) of 1.36 eV and 1.16 eV, respectively (the normalized fitting parameters are shown in Supplementary Table 3). In the one-pot spectrum, a peak at 286.9 eV and FWHM of 1.63 eV are indicated as the normalized values (Supplementary Table 3). We recognize that this component in the one-pot process is different from the other two components. Inamura et al.¹⁹ previously reported that the isopropylamine (iPA_m) molecule attached to a Ni catalyst strain-free exhibits a peak at ~285 eV in the C 1s spectrum below 300 K. We presume that this chemical component can be assigned to the two components at 286.4 eV from the sequential and one-pot (IPA) processes. The difference between the FWHM in our observation (1.36 eV and 1.16 eV) and the previously reported value is less than 2.2 eV, and it can be influenced by the states of isolation as an ordered phase at grain boundaries in perovskite films and strain-free as disordered phase of iPA_m molecules or iPA_mH⁺ cations. The peak at 286.9 eV in the one-pot spectrum can be assigned to the C–N bond of the complexing methylamine molecule¹⁷. In the analysis of the N 1s spectra in Fig. 3g–i, the three perovskite films each indicate a major peak at ~401.0 eV, attributed to FA, and a minor peak at ~402.0 eV, assigned to MA. We note that sequential and one-pot (IPA) spectra have indicated one more component at ~400 eV with FWHM (full-width at half maximum) of ~2.0 eV (Fig. 3g) than that of one-pot (Fig. 3i). This peak corresponds to the amino group of iPA_m¹⁹.

The potential presence of iPA_mH⁺ or analogous compounds in the three perovskite films annealed at 150 °C after deposition by a sequential, one-pot and one-pot (IPA) process was also confirmed by Fourier transform nuclear magnetic resonance (FTNMR) spectroscopy (Supplementary Fig. 10). For the NMR measurements, a bottle containing MACl (40 mol%), PbI_2 (1.6 M) and IPA (20 mol%) in deuteriated *N,N*-dimethylformamide ($[\text{D}_7]$ DMF) was sealed and set in the temperature range of 75–85 °C for a reaction time of approximately 12 h. The ¹H FTNMR spectra of three precursor solutions, namely IPA– PbI_2 , iPA_mHCl– PbI_2 and MACl– PbI_2 , dissolved in $[\text{D}_7]$ DMF, were compared with those of MACl–IPA– PbI_2 before and after heating at 80 °C (Supplementary Fig. 11). Here, note that MACl did not dissolve well without PbI_2 in DMF. Interestingly, the ¹H NMR peak (5.75 ppm) due to the OH of IPA disappeared in the solution of MACl–IPA– PbI_2 , even before prolonged heating. This means that when MACl is added to IPA– PbI_2 , an unspecified

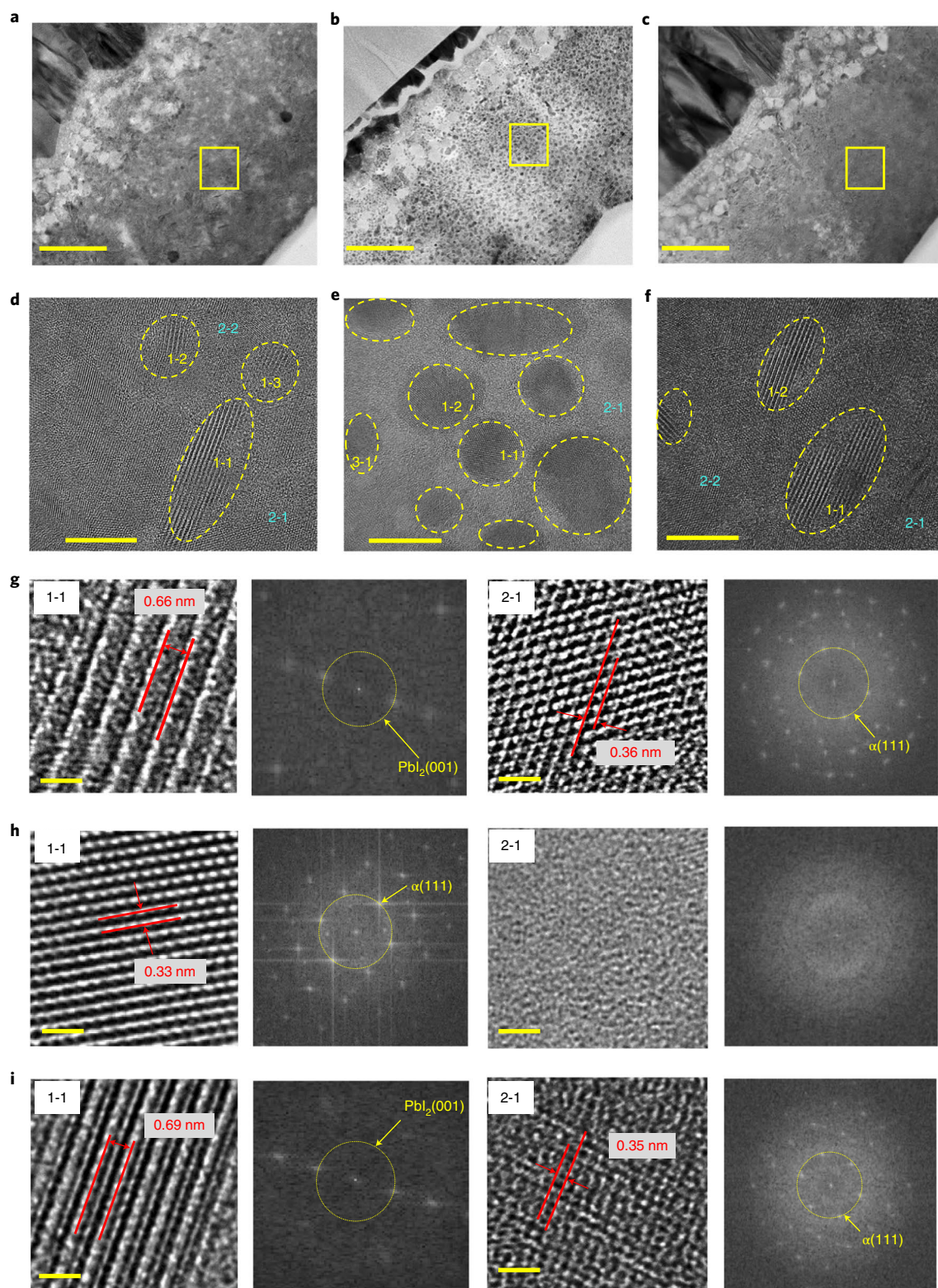


Fig. 2 | HR-TEM characterization of the FAPbI₃ films. **a–c**, Cross-sectional HR-TEM images of the FAPbI₃ films prepared by the sequential (**a**), one-pot (**b**) and one-pot (IPA) (**c**) processes (scale bars, 200 nm). **d–f**, Magnifications of the areas in **a–c** marked by the solid-line squares for the sequential (**d**), one-pot (**e**) and one-pot (IPA) (**f**) processes (scale bars, 10 nm). The numbers in **d–f** correspond to the crystals for which IS values and FFT patterns were determined. Two areas are represented: inside (1) and outside (2) the yellow dashed-line circles/ellipses. Crystallite FAPbI₃ grains are present inside the dashed-line circles/ellipses. **g–i**, Magnified HR-TEM images with IS values and FFT patterns of two representative areas: inner (1-1) and outer (2-1) regions on the sequential (**g**), one-pot (**h**) and one-pot (IPA) (**i**) perovskite films (scale bars, 1 nm). The red solid lines and red numbers represent lattice planes and the corresponding IS value, respectively.

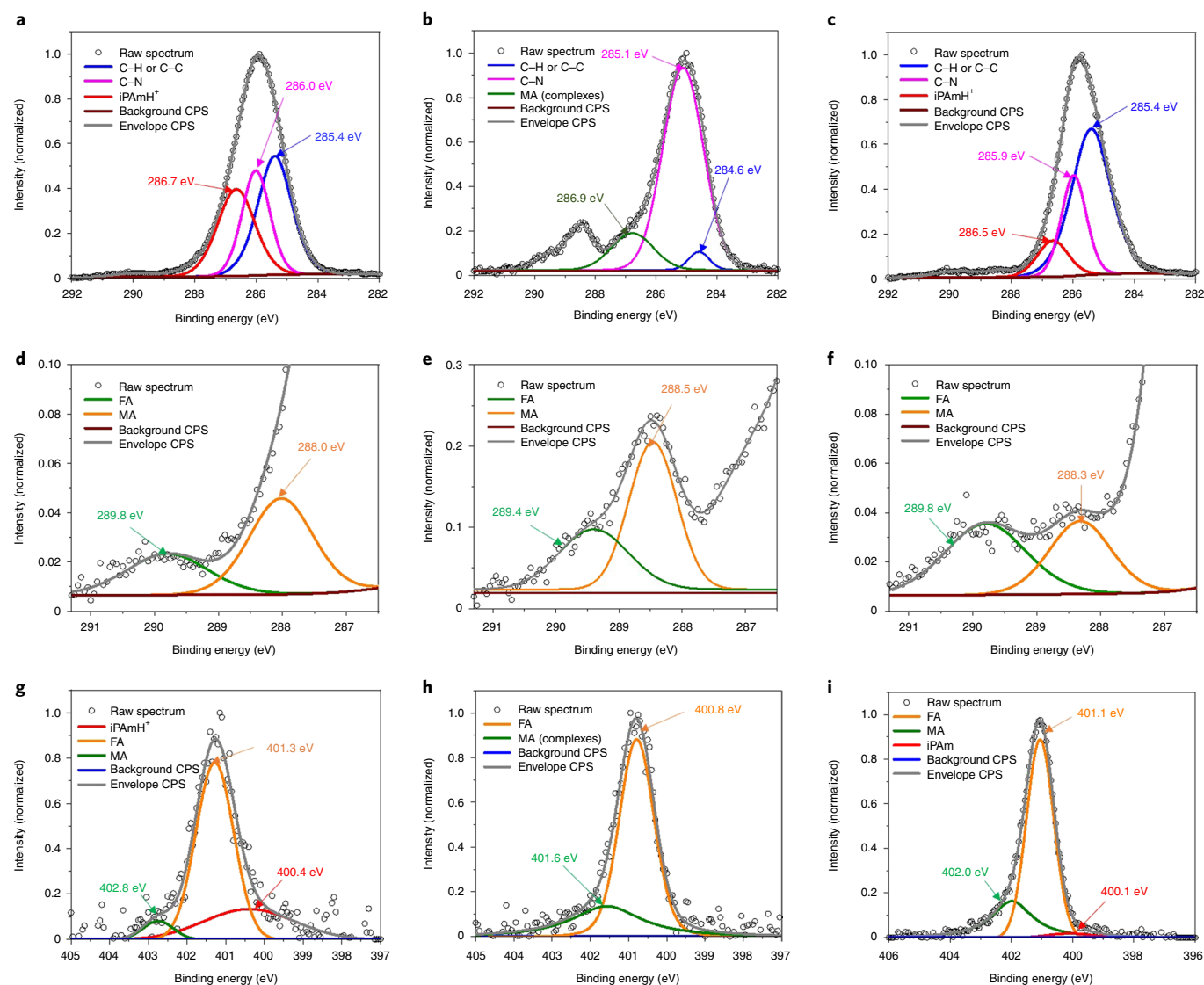


Fig. 3 | HR-XPS chemical analysis of the three FAPbI₃ perovskite thin films. a–c, The deconvoluted C 1s HR-XPS spectra of the FAPbI₃ films prepared by the sequential (a), one-pot (b) and one-pot (IPA) (c) processes. **d–f**, Magnification of the C 1s spectra of films from the sequential (d), one-pot (e) and one-pot (IPA) (f) processes in the region between 290.7 and 287.5 eV. **g–i**, The deconvoluted N 1s HR-XPS spectra of the FAPbI₃ films prepared by the sequential (g), one-pot (h) and one-pot (IPA) (i) processes. The optimized annealing times for the FAPbI₃ films were 15 min for the one-pot and one-pot (IPA) processes and 30 min for the sequential process. The fitted parameters are summarized in Supplementary Table 3.

intermediate phase forms with the iodoplumbate (multivalent PbI₂) dissolved in solution. In particular, the MACI-IPA-PbI₂ precursor solution after heating showed new but very small peaks (3.64 and 1.37 ppm) at the same positions as the NMR chemical shifts found in the spectrum of iPA₄HCl. This transamination to iPA₄H⁺ when MACI and IPA coexist in lead halide perovskite was further investigated by Fourier transform infrared (FTIR) analysis. In the FTIR spectra of the PbI₂, MACI-PbI₂, IPA-MACI-PbI₂ and iPA₄HCl-PbI₂ precursor solutions (Supplementary Fig. 12a), we found a broad absorption peak at 1,020 cm⁻¹ in only the MACI-IPA-PbI₂ and iPA₄HCl-PbI₂ spectra. To assign this peak, we additionally conducted FTIR measurements on MACI (crystal), iPA₄HCl (crystal) and IPA (liquid) as reference compounds (Supplementary Fig. 12b). The peak at 1,020 cm⁻¹ in the iPA₄HCl (crystal) spectrum is the same as that in the spectra of the IPA-MACI-PbI₂ and iPA₄HCl-PbI₂ precursor solutions. The peak at 1,000 cm⁻¹ for MACI (crystal) also appears in the spectrum of the IPA-MACI-PbI₂ precursor solution. The broad absorption peak seen for the MACI-PbI₂

solution can also be considered to contain the 1,000 cm⁻¹ peak. This implies that complete transamination did not occur in the MACI-IPA-PbI₂ precursor solution during heating. Nevertheless, the FTIR results provide strong evidence for the formation of iPA₄HCl or its analogues in the precursor solution. Furthermore, in recent studies, the decomposition of organic hybrid perovskites under thermal stress, yielding ammonium (NH₄⁺) and dimethylammonium ((CH₃)₂NH₂⁺) by deprotonation or methyl transfer from methylammonium (CH₃NH₃⁺)²⁰, and the alkylation of amines with alcohols^{21,22} have been discussed. In these studies, the transamination of IPA^{23–25} might have been possible through the partial secondary chemical reaction of a mixture comprising MACI, IPA, PbI₂ and partially ionized chemical species (for example, PbI₆ⁿ⁻, Pbⁿ⁺ (ref. 26), halide induced compounds (for nucleophilic amination)²⁷, with residual DMF and dimethylsulfoxide (DMSO) used as solvents during annealing of the perovskite film at 150 °C (low-temperature amination^{22,28}) under a remarkably high uniaxial strain of a few megapascals (the elastic constant of FAPbI₃ is in the gigapascal

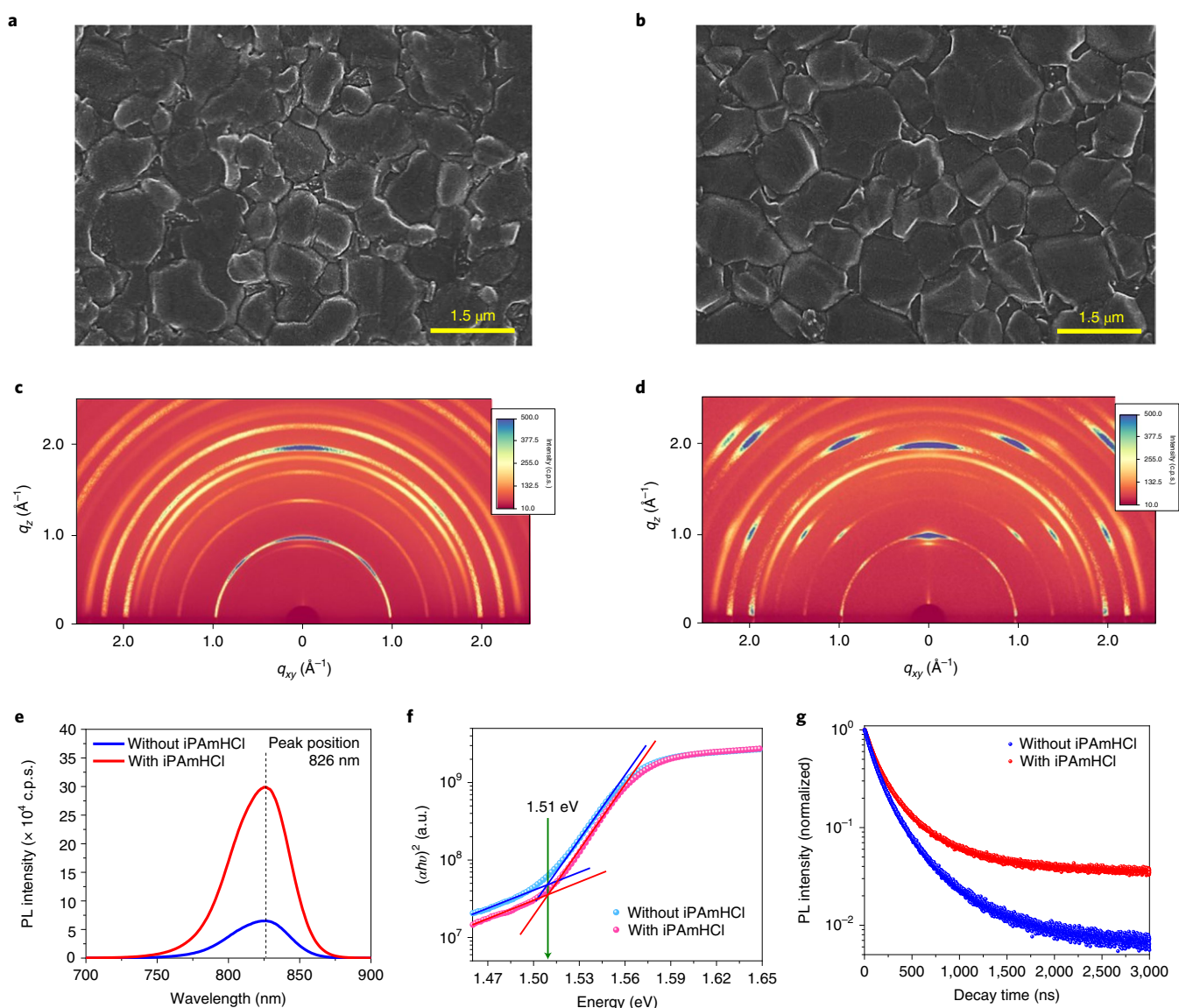


Fig. 4 | Surface, crystal and optophysical properties. **a, b**, Field-emission scanning electron microscopy surface morphologies of the perovskite films deposited without **(a)** and with **(b)** iPAHCl (15 mol%) on mp-TiO₂ substrate. **c, d**, Comparison of the 2D GI-WAXD patterns of the perovskite films deposited without **(c)** and with **(d)** iPAHCl (15 mol%) incorporation. q_{xy} , in-plane scattering direction; q_z out-of-plane scattering direction. **e–g**, The optophysical properties of the films are illustrated by their PL spectra **(e)**, Tauc plots **(f)** and TRPL decay curves **(g)**. The optical bandgaps of the two perovskite films were determined from the Tauc plots from the intercept of the extrapolated linear fits of the $(\alpha h\nu)^2$ versus energy plots near the absorption edge, where α is absorption coefficient, h is Planck's constant and ν is frequency. c.p.s., counts per second. The TRPL curves were fitted by the biexponential function $y = a_1 \exp(-t/\tau_1) + a_2 \exp(-t/\tau_2)$, where τ_1 and τ_2 denote the fast and slow decay time constants, which correspond to the radiative and non-radiative lifetimes, respectively, y is the offset, t is the time constant and a_n is the amplitude constant (where n is 1 or 2).

range)²⁹ from pre-annealing conditions (Fig. 1a–c). Considering several peculiarities observed in halide perovskites, we propose that it may be possible to generate iPAH⁺ molecules in the sequential and one-pot (IPA) processes. Although we cannot explain the precise chemical reaction pathway, we believe that the FAPbI₃ obtained by the IPA processes is likely to contain residual organic cations such as iPAH⁺ in addition to FA.

Performance of iPAHCl-stabilized FAPbI₃ PSCs

Based on these findings, we conducted model experiments to fabricate PSCs by adding 15 mol% iPAHCl to the one-pot solution for the preparation of the FAPbI₃ layer. Further details are provided in the Methods. As shown in Fig. 4a,b, the grain size in the FAPbI₃ film slightly increased when 15 mol% of iPAHCl was added.

Figure 4c,d presents the 2D GI-WAXD patterns for the perovskite thin films deposited on mesoporous TiO₂ (mp-TiO₂) without and with iPAHCl. Peaks attributed to the δ -FAPbI₃ phase, PbI₂ and other species can barely be observed (Supplementary Fig. 13a), only those assigned to α -FAPbI₃. In particular, the peaks of the 2D structures produced by iPAH⁺ cations larger than those of FA were not visible at low angles. The peak at 13.92° is in full agreement with the (001) peak for single-crystal α -FAPbI₃ (ref. 30). In addition, as shown in Supplementary Fig. 13b, we found that the diffraction intensity increased as the number of iPAH⁺ cations increased. This increase in intensity can be attributed to increased crystallinity or crystal orientation in a particular direction. Because the half-widths of the diffraction peaks were almost the same, the addition of iPAHCl did not appear to increase the FAPbI₃ crystal

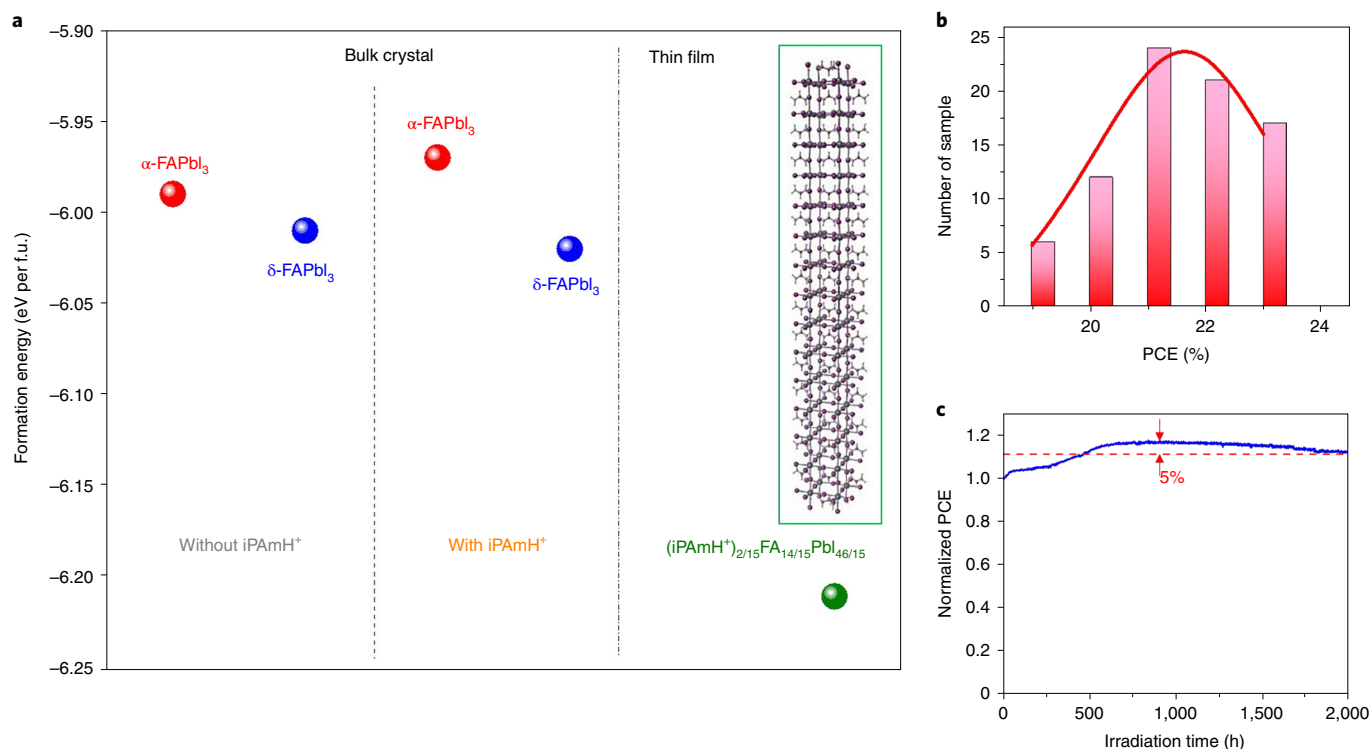


Fig. 5 | Computational investigation and PSC performance. **a**, Computational investigation of the stabilization of the α -FAPbI₃ crystal by the iPAHCl molecule located near grain boundaries. Simulated formation energies for α -FAPbI₃ and δ -FAPbI₃ crystals in the bulk and thin-film phases. An iPAH⁺ cation located at the grain boundary of the α -FAPbI₃ supercell was simulated and found to considerably reduce the formation energy compared with the δ -FAPbI₃ crystal. The formation energies of five different concentrations were compared: the α - (red circle) and δ -phases (blue circle) of bulk FAPbI₃, and the bulk α - (red circle), δ - (blue circle) and thin-film phases (green circle) of (iPAH⁺ + FA)PbI₃. The thin-film configuration of (iPAH⁺ + FA)PbI₃ is the most stable in terms of free energy. In the right panel, the atomic structure of the thin-film configuration is presented. Inset: simulated model of the super cell (iPAH⁺)_{2/15}FA_{14/15}PbI_{46/15}. f.u., formula unit. **b**, PCE histograms of 80 different PSCs fabricated with perovskite films incorporating 15 mol% iPAHCl. The red curve shows the frequency distribution of each efficiency. **c**, Evolution of the PCE for a representative solar cell, measured over 2,000 h under 1-Sun irradiation and dynamic *J*-*V* scanning (repeated sweep every 84 min). The initial PCE and the final PCE after 2,000 h were 19.66 and 22.01%, respectively. The red dashed line indicates the normalized PCE after 2,000 h.

size. Thus, the addition of iPAHCl to FAPbI₃ with MAcl led to the crystallization of pure α -FAPbI₃ and the preferred orientation of (100)_o, which was also confirmed by GI-WAXD measurement (Fig. 4c,d and Supplementary Fig. 14). This result is attributed to the contribution of the iPAH⁺ cation to the van der Waals interaction responsible for crystal orientation, similarly to the cases in which organic cations with longer alkyl chains, such as butylammonium and octylammonium, were added³¹. Moreover, from a comparison of the 2D GI-WAXD patterns of the sequential, one-pot (IPA) and iPAHCl-incorporated FAPbI₃ films (Figs. 1a,c and 4d), it is believed that iPAH⁺ contributes importantly to reducing the formation of the δ -, 4H- and 6H-FAPbI₃ phases. For the system in which iPAHCl has been added, iPAHCl may partially disappear during the heat treatment process owing to the volatile Cl⁻ ions, as occurs on addition of MAcl. However, based on the results of an NMR analysis (Supplementary Fig. 15), it is believed that a substantial amount of iPAHCl still remains in the perovskite layer. The residual iPAHCl in the perovskite layer is present mainly at the FAPbI₃ grain boundaries because no shift in the bandgap or diffraction angle of α -FAPbI₃ was observed after its addition. Furthermore, Cl⁻ ions were not detected on the surfaces of either perovskite film (Supplementary Fig. 16). By contrast, in the depth profile analysis of the perovskite layer by ToF-SIMS (Supplementary Fig. 17), we observed that the distribution of Cl⁻ ions is higher in the nanostructured TiO₂ electrode. This (as will be seen later) is one of the main reasons for the excellent stability observed under long-term

light irradiation, including ultraviolet irradiation, even when TiO₂ is used as the photoelectrode³². Figure 4e shows the photoluminescence (PL) measured for the FAPbI₃ thin film without and with iPAHCl (15 mol%) incorporation, deposited on a glass substrate. The wavelength of the maximum PL peak did not change, although the intensity notably increased on addition of iPAHCl. In addition, the iPAHCl-incorporated perovskite contains a PL peak from pseudo-cubic iPAHCl-FAPbI₃ (Supplementary Fig. 18). The maximum peak of the PL coincides with the bandgap determined from the Tauc plot (Fig. 4f), obtained from ultraviolet/visible (UV/Vis) spectroscopic measurement. This confirms that the addition of the iPAH⁺ cation has a negligible effect on the macroscopic bandgap (1.51 eV). Figure 4g shows the time-resolved PL (TRPL) decay curves obtained by the time-correlated single-photon counting technique from the same thin films as used for the previous PL measurement. As shown in Fig. 4g and Supplementary Fig. 19, the radiative and non-radiative charge decay lifetimes were longer for the perovskite layer deposited after adding 15 mol% iPAHCl, implying a low defect concentration in the α -FAPbI₃ crystals. Accordingly, we conclude that the added iPAH⁺ cations contribute to the reduction in defect concentration through increased crystal orientation along the (100) plane with pure α -FAPbI₃ as a single crystal without the formation of 2D perovskite or substitution by FA, as shown in Fig. 4d. In particular, the addition of iPAHCl appears to contribute to the stabilization of the α -phase of FAPbI₃, allowing the fabrication of PSCs with favourable photocurrent density

because the inherent bandgap of FAPbI₃ was not widened. Although α -FAPbI₃ can be stabilized by alloying^{1,2,4} and the formation of 2D perovskite at grain boundaries^{33,34}, we believe that iPAHCl added to the precursor solution stabilizes the α -phase in a number of complex ways, including the surface passivation of FAPbI₃ crystals, as reported previously^{9,33,34} and evidenced by our computational simulation (Fig. 5a), and the induction of single crystal-like grains with few defects³⁰.

The addition of 15 mol% iPAHCl to the precursor solution resulted in a low defect concentration, thereby stabilizing the α -FAPbI₃ phase. Therefore, we fabricated a series of PSCs under the same specifications and investigated their performance. Figure 5b shows the histograms of the 80 PSCs fabricated, and the average efficiency is higher than 21%. The current density–voltage (J – V) curves for a representative PSC, measured with and without anti-reflection polymer on the glass substrate, are shown in Supplementary Fig. 20a. The corresponding external quantum efficiencies of the PSC fabricated with the incorporation of iPAHCl are displayed in Supplementary Fig. 20b, and it can be noted that the integrated current density ($J_{sc} = q/hc \int \lambda \Phi(\lambda) EQE(\lambda) d\lambda$), where c is the speed of the light, λ is the wavelength, q is elemental charge, Φ is the solar spectrum of AM1.5G and EQE is external quantum efficiency, agrees with the J_{sc} value from the J – V curve. One of the devices with the highest efficiencies among those fabricated under the same conditions was certified to have an efficiency of 23.9% in an accredited laboratory (Newport, USA), based on the quasi-steady-state method (Supplementary Figs. 21 and 22). The long-term operation of the PSC was investigated by employing the hallmark J – V measurement long-term stability test³⁵. In this test, J – V curves were measured by scanning one cycle every 84 min under 1-Sun irradiation in nitrogen atmosphere at room temperature for 2,000 h, without a metal mask using a class A solar simulator. Figure 5c shows the long-term behaviour of the normalized efficiency of the as-fabricated device. The efficiencies measured accurately with a metal mask and class 3A solar simulator before and after long-term operation are shown in Supplementary Fig. 23. The efficiency gradually increased up to approximately 1,000 h, and then gradually decreased in the following 1,000 h. We surmise that the initial increase in efficiency of the device reflects the length of time required to optimize the surface and internal structure for charge transfer by the slow redistribution of iPAH⁺ near the FAPbI₃ grain boundary (Supplementary Fig. 24). The decrease in efficiency after achieving the maximum efficiency can be attributed to the degradation of the perovskite surface and interface owing to the diffusion of the 4-*tert*-butylpyridine used as an additive in the organic hole conductor. Another factor is the corrosion of the gold layer used as a counter electrode as a result of iodide migration in FAPbI₃, with the transformation of the FAPbI₃ α -phase to the δ -phase³⁶. For reference (Supplementary Fig. 25), in a phenylethylammonium iodide-stabilized PSC³³, the initial PCE of 22.02% rapidly decreased during the long-term J – V test to 60% of the initial value after approximately 125 h. Therefore, we believe that the relatively good long-term stability achieved following the addition of iPAHCl is due to multiple factors, including the stabilization of the α -phase by the presence of iPAH⁺ cations on the surface of FAPbI₃ and the interfacial stabilization of the perovskites by Cl[−] ions on the surface of the TiO₂ electrodes.

Conclusions

A comparison of the FAPbI₃ perovskite films fabricated with MACl and IPA in the presence of iPAHCl by one-pot and sequential solution processes suggested the possible incorporation of iPAH⁺ cations into FAPbI₃. Based on these findings, the effect of the addition of iPAHCl to FAPbI₃ was systematically investigated. When 15 mol% iPAHCl was added to FAPbI₃ in a PSC device, the certified efficiency was 23.9%. Moreover, the initial efficiency of the as-fabricated device was retained, even after 2,000 h of exposure

to 1-Sun illumination. These enhanced PSC device characteristics can be attributed to the impact of iPAHCl, which acts to stabilize the α -phase, reduce defects by surface passivation of FAPbI₃ and increase the preferred crystal orientation of single crystals.

Methods

Materials. Unless otherwise stated, all materials were purchased from Sharechem, Sigma-Aldrich, Alfa Aesar, Junsei Organics and TCI, and were used as received.

Fabrication of solar cells. A dense blocking layer of TiO₂ (approximately 60 nm in thickness, bl-TiO₂) was deposited onto an F-doped SnO₂ (FTO, Pilkington glass, TEC-8, 2.2T) substrate by spray pyrolysis using titanium diisopropoxide bis(acetylacetonate) in ethanol (1:10, v/v) at 450 °C. A 150-nm-thick layer of nanostructured TiO₂ was spin-coated at 1,500 r.p.m. onto the bl-TiO₂/FTO substrate using TiO₂ paste (Sharechem) diluted with 2-methoxyethanol and then annealed at 500 °C for 1 h in air to remove the organic components. In the sequential process, PbI₂ (1.30 M) was dissolved in a 1 ml mixture of DMF and DMSO (8:2, v/v) at 80 °C. The dripping solution was prepared by dissolving formamidinium iodide (FAI; 80 mg) in 1 ml IPA containing 10 mg MACl. The PbI₂ precursor solution was spin-coated on the mp-TiO₂ substrate at 3,000 r.p.m. for 30 s to obtain a transparent PbI₂–DMSO intermediate film. Subsequently, 0.35 ml of the prepared dripping solution that included FAI and MACl was dripped on top of the PbI₂–DMSO intermediate film and spin-coated at 5,000 r.p.m. for 30 s. The colour of the film changed to dark brown during spin-coating, and the film was then dried on a hot plate at 100 °C for 30 min and then at 150 °C for 30 min. The heat-treated film was washed with IPA and annealed at 100 °C for 30 min. For the one-pot and one-pot (IPA) processes, a 1.6 M precursor solution of FAPbI₃ and 40 mol% MACl in DMF and DMSO (6:1, v/v) was prepared, respectively. For the one-pot (IPA) process, IPA (20 mol%) was additionally added into the one-pot precursor solution. To modify the optimized coating-protocol for the formation of FAPbI₃ films incorporating iPAHCl, 15 mol% iPAHCl was added to the precursor solution containing 40 mol% MACl and 1.6 M FAPbI₃ (homemade). Then, the precursor solution was coated onto the nanostructured TiO₂/bl-TiO₂/FTO substrate at 1,000 r.p.m. for 30 s and 5,000 r.p.m. for 10 s with dripping with chlorobenzene or diethyl ether, respectively. The coated FAPbI₃ films were annealed on a hot plate at 150 °C for 10 min. We used an organic hole conductor prepared with a solution of 86 mg Spiro-MeOTAD (EM Index) in 1 ml chlorobenzene combined with 20 μ l lithium bis(trifluoromethanesulfonyl)imide (TFSI)/acetonitrile (500 mg/1 ml), FK 209 Co(III) TFSI/acetonitrile (230 mg/1 ml) and 35 μ l 4-*tert*-butylpyridine, which was spin-coated on the perovskite layer/mp-TiO₂/bl-TiO₂/FTO substrate at 3,000 r.p.m. for 30 s. Finally, a gold counter electrode was deposited by thermal evaporation; the active area of this electrode was fixed at 0.16 cm².

Crystallographic investigation. The GI-WAXD measurements were performed at the PLS-II 6D UNIST-PAL beamline of the Pohang Accelerator Laboratory (PAL), Korea. The X-rays from the bending magnet were monochromatized to 18.986 keV ($\lambda = 0.6530$ Å) using a double-crystal monochromator that was focused both horizontally (H) and vertically (V) (90 (V) $\times$ 120 (H) μ m² FWHM) on the sample position using a sagittal Si(111) crystal and toroidal mirror, respectively. The GI-WAXD sample chamber was equipped with a five-axis motorized stage for fine sample alignment, and the 2D GI-WAXD patterns were recorded with a 2D CCD detector (MX 225-HS, Rayonix). The diffraction angles were calibrated using NIST SRM660b, lanthanum hexaboride of approximately 6g (LaB₆) and the sample-to-detector distance was set at 246.4 mm. An incidence angle of 0.80° was used for the in situ GI-WAXD experiments to obtain crystallographic information on the overall thickness of the perovskite films and ensure that all incoming X-rays were directed onto the small length of the perovskite film sample without being blocked by the substrate or sample stage or just passing by. Various incidence angles in the range 0.117–0.8° were used for the static GI-WAXD experiments on the sequential, one-pot, one-pot (IPA) and one-pot (no MACl) samples before and after heat treatment, as well as on the FAPbI₃ film with or without iPAHCl incorporation. For the static GI-WAXD experiments, an exposure time of 30 s was used at each incident angle. In situ GI-WAXD patterns were obtained every 5 s for the first 10 min, and then every 1 min between 10 and 30 min, with an X-ray exposure time of 3.5 s each. It took 2 min to raise the temperature of the FAPbI₃ film from room temperature to 150 °C, and then the temperature was maintained at 150 °C for 30 min using a fine-tuned temperature controller (Eurotherm 3504 Temperature Controller, Eurotherm). The X-ray diffraction patterns of the iPAHCl-incorporated FAPbI₃ films were measured in the scattering angle range of $2\theta = 5$ –40° using a SmartLab X-ray diffractometer (Rigaku) with a copper target operated at 45 kV and 200 mA ($K_{\alpha 1}$ optics monochromator, symmetric Johansson Ge(111) curved optic crystal, vertical θ/θ geometry with a horizontal sample mounting geometry and high-speed 1D (D/tEX Ultra 250) detector) at the UNIST Central Research Facilities.

Observation of surface morphologies and crystal properties. The scanning surface morphologies of the FAPbI₃ thin films were observed at $\times 80K$ and

×5K magnifications by cold field-emission scanning electron microscopy (FE-SEM; SU-8220, Hitachi) with a beam power of 5 kV. To investigate the local crystal properties, most samples were prepared using a focused ion beam (FIB; Helios NanoLab 450 model, FEI) and analysed by FE-SEM. The FIB milling/polishing process ensured that the samples had a smooth surface. Furthermore, the FIB-based milling process enabled the sample to be thinned to 100 nm, as required for HR-TEM characterization. HR-TEM images were collected using a JEM-2100F microscope (JEOL). ImageJ v.1.52a and Gatan Digital Micrograph v.3.9 and Gatan Microscopy Suite v.3.4.3 were used to analyse the HR-TEM images.

Measurement of optical properties. To measure the optical absorption and bandgaps, the prepared FAPbI₃ samples were scanned using a V-780 UV/Vis/NIR spectrophotometer (Jasco). The system comprised a Czerny–Tuner mount, a double monochromator, a fully symmetrical double beam (deuterium (spectral range 190–350 nm) and halogen lamps (spectral range 330–1600 nm)) and a photomultiplier tube with a Peltier-cooled InGaAs photodiode. The scan range was 190–1,600 nm with spectral bandwidths of 0.5 and 2.0 nm in the UV/Vis and near-infrared (NIR) regions, respectively; the photometric ranges were ~4–4 Abs for UV/VIS and ~3–3 Abs for NIR. The photometric accuracies were ±0.0015 Abs (0.0–0.5 Abs), ±0.0025 Abs (0.5–1.0 Abs) and ±0.3%T. The root-mean-square noise was 0.00003 Abs.

FTIR investigation. The FTIR spectra of the perovskite precursors and powders were recorded using a Frontier spectrometer (Series 6800, PerkinElmer) with best-resolution parameters and 20 accumulations in attenuated total reflection mode. The instrument specifications consisted of two temperature-stabilized deuterated triglycine sulfate detectors (one optimized for mid-infrared (MIR) with an enhanced signal-to-noise ratio (SNR) of 15,000:1 peak-to-peak for a 5-s scan, and one suited to NIR with <10 µA root-mean-square noise over a range of 250 nm for a 1 min scan), an optical system with a KBr beam splitter for MIR for the spectral range 8,300–350 cm⁻¹ and a best resolution of 0.4 cm⁻¹, a CaF₂ beam splitter for the spectral range 14,700–2,000 cm⁻¹ and a best resolution of 0.5 cm⁻¹, a standard atmospheric vapour compensation/automatic performance verification/absolute virtual instrument, standard gold optics and standard KBr windows.

FTNMR investigation. For NMR analysis, the perovskite films were dissolved in [D₂]DMF. The FTNMR investigation was conducted using a VNMR5 600 spectrometer (magnet 14.09 T, 600 MHz).

Charge carrier dynamics investigation. Steady-state PL spectroscopy and time-correlated single-photon counting (TCSPC) were performed using a FluoTime 300 spectrometer (PicoQuant) equipped with a PMA-C-192-M detector, a high-resolution excitation monochromator and a hybrid photomultiplier detector assembly (PMA Hybrid 40, PicoQuant). The FAPbI₃ samples were installed in an integrating sphere and excited at 512 nm. The excitation fluence intensities were set at 1.5 mW cm⁻², measured using a power meter (Thorlabs PM100D). The diode laser was controlled by a PDL 820 driver to control the picosecond pulses of the excitation light source (internal repetition rates between 196 and 80 MHz), and maintained at 10 MHz during TCSPC scanning. The laser beam was incident on the backside (FAPbI₃ side) of the FAPbI₃ film/glass substrate. TCSPC was performed at the wavelength of maximum PL intensity. The FAPbI₃ samples were coated on quartz glass substrates.

ToF-SIMS measurement. ToF-SIMS profiling, imaging and 3D tomography were performed to measure the lateral and depth distributions of the perovskite layers and completed devices. The samples were analysed using a ToF-SIMS V spectrometer (IONTOF) in high-current bounce mode (double crossover of beam alignment in the ion lens, 21 ns and 8 pA). The hybrid system consisted of a basic configuration consisting of a ToF-SIMS generation IV sputter gun and detector. The Bi⁺ primary beam was set at 25 keV and 1.2 pA. The sputter beam conditions were set at 1.0 KeV and 35 nA for Cs⁺ for elemental detection at negative polarity, and at 500 eV and 100 nA for O₂ for elemental detection at a non-interlaced positive polarity (1 frame, 2 s pause, 2 s sputter). The time of flight of pulse length was 10.9 ns, the repetition rate was 10 kHz, the acceleration voltage of the secondary ions was 2,000 V, the extraction timing was 0 s, the flight length was 2 m, the ion mirror voltage was 20 V, the acquisition time was 500 s, the incident angle was 45°, the sputter size was 200 µm × 200 µm, the analysis area was 50 µm × 50 µm and the primary ion dose density was 2.04522e⁺ (ref. ¹⁴).

Investigation of the energy levels and chemical analysis. XPS measurements were conducted in an ultrahigh vacuum using a Scienta 2002 analyser at the PLS-II 10A2 beamline of PAL. To apply a bias potential, a tantalum sheet was put in direct contact with the FAPbI₃ film. The valence bands of the samples were measured at 130 eV. A photon energy of 700 eV was used for the core level measurement of N 1s. iPAmH⁺ or iPAm incorporation was analysed by measurement of C 1s at 900 eV. The measurements were conducted in a vacuum chamber at 10⁻⁹–10⁻¹⁰ torr, and three sweeping scans were collected.

Evaluation of fabricated FAPbI₃ solar cells. The *J*–*V* curves were obtained with a source meter (Keithley 2420) using a solar simulator (Newport, Oriel Class AAA, 94043A) with 100 mA cm⁻² illumination (AM 1.5G) and a calibrated silicon reference cell (Newport, Model 91150V) certified by National Renewable Energy Laboratory (NREL). The calibrated 1-Sun illumination was sustained automatically using an exposure control instrument (Newport, Model 68951). The *J*–*V* curves were measured in reverse scan mode. The step voltage and delay time were fixed at 10 mV and 40 ms, respectively. The delay time was set at each voltage step before measuring the current. The *J*–*V* curves of the devices were measured by masking the active area with a metal mask (active areas of 0.096 cm² (small cell)). The integrated current densities of PSCs were evaluated using external quantum efficiency system (IQE-200B, Newport, Oriel), which can monitor the range from 350 nm to 1100 nm in high resolution as 0.1 Hz (4–100 Hz chopper frequency), monochromator band width as 1–30 nm FWHM, wavelength accuracy ±0.5 nm under 100 W xenon lamp. The steady-state current densities of the solar cell devices were measured under N₂ gas using an electrochemical station (Autolab PGSTAT302N, Metrohm Autolab). The dynamic *J*–*V* scanning long-term stability of the encapsulated PSC devices was investigated under N₂ gas, with one scan cycle every 84 min under 1-Sun irradiation. This test chamber was composed of stainless steel with a quartz glass window in front of the light source. The gas composition in the chamber was controlled using a vacuum pump and gas injection nozzles. The temperature in the loading stage was controlled by water flowing through the inlet pipeline from an external circulating water bath (RW3-0525, Lab Companion). The *J*–*V* curves were measured with a source meter (Keithley 2450) using a solar simulator (Newport, Oriel Class A(ABB), IEC 60904-9 EDITION2) at 100 mA cm⁻² illumination (AM 1.5G) and a calibrated silicon reference cell (Newport, Model 91150V) certified by NREL. The calibrated 1-Sun illumination condition was sustained automatically using an exposure control instrument (Newport, Model 68951).

Density functional theory simulation. Density functional theory calculations were performed to investigate the phase stability of the iPAmHCl-incorporated FAPbI₃. We modelled supercell structures of the α-, δ- and thin-film phases accommodating 16 formula units. In each supercell of the α- and δ-phases, one iPAm molecule was substituted for one FA molecule at the A site, while two iPAm molecules were attached to the surfaces of the slab model, iPAmH⁺FA₁₄PbI₁₅I₄₆. To optimize the given supercell structures, we used a plane wave basis set with a cut-off energy of 600 eV within the projector augmented wave method³⁷ implemented in the Vienna Ab initio Simulation Package³⁸. To embody the correlation and exchange terms of the electron–electron interactions, we employed the generalized gradient approximation within the Perdew–Burke–Ernzerhof formalism³⁹. Van der Waals interactions were considered according to the Tkatchenko–Scheffler method⁴⁰. The internal coordinates as well as the lattice parameters were relaxed until the minimum force became less than 0.03 eV Å⁻¹. For *k*-point sampling, we used a regular grid of 3 × 4 × 3 for bulk supercells and 8 × 8 × 1 for the thin-film configuration. The rotational entropy of the FA molecule was evaluated on the basis of the moment of inertia reported previously⁴¹.

Data availability

All data generated or analysed during this study are included in the published article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

B.-w.P., H.W.K. and S.I.S. conceived this work. B.-w.P. and H.W.K. prepared the perovskite thin layers and solar cells and interpreted all data from the analyses. Y.L. and D.Y.L. fabricated the materials and devices for the perovskite solar cells. G.K. and K.-j.K. conducted the XPS investigation. M.G.K. performed the chemical analysis. Y.K.K. carried out HR-TEM analysis. J.I. calculated the formation energy of iPAmHCl-FAPbI_3 . T.J.S. guided the XPS and GI-WAXD investigations and contributed to their interpretation. B.-w.P. and S.I.S. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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